

DEVASHISH BOSE

Machine Learning — Quantitative Research — Data Analysis — Python — AI/ML
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Professional Summary

Machine Learning engineer and quantitative analyst with hands-on experience in **Python, statistical analysis, ML model development, and data-driven research**. Built production-grade ML pipelines achieving sub-200ms latency and applied **K-Means clustering, RFM analysis, and predictive modeling** on 500K+ transaction datasets. Experienced in **LLM post-training (RLHF/SFT), RAG pipelines, time-series analysis, and AI-driven systems**. Strong analytical thinking, attention to detail, and a passion for applying ML to quantitative research and algorithmic decision-making. **Immediate joiner — open to relocating to Jodhpur (Onsite)**.

Technical Skills

Machine Learning & Statistics: Python, Scikit-Learn, Pandas, NumPy, Statistical Analysis, Time-Series Analysis, K-Means Clustering, RFM Analysis, Predictive Modeling, Feature Engineering, A/B Testing, Regression, Classification, Random Forest, LSTM

Quantitative Research & Analytics: Data-Driven Research, Market Analysis, Risk Modelling, Pattern Recognition, Churn Prediction, Business Intelligence, KPI Analysis, EDA, ETL Pipelines

AI & LLMs: RLHF, SFT, Prompt Engineering, Prompt Optimization, LangChain, HuggingFace, Transformers, RAG Pipelines, FAISS, ChromaDB, Pinecone, AI Data Annotation, Content Evaluation

Big Data & Tools: Apache Spark, PySpark, Hadoop, Hive, MapReduce

Visualization & BI: Power BI, Tableau, Matplotlib, Seaborn, Plotly, Excel

Databases & Backend: PostgreSQL, MySQL, MongoDB, FastAPI, Flask, Streamlit, REST APIs, Git, GitHub, Linux, VS Code, Jupyter

Languages: Python, SQL, R, Bash, C++, JavaScript

Experience

LLM Post-Training Intern Jul 2025 – May 2026

Ethara AI (Remote)

- Executed **RLHF and Supervised Fine-Tuning (SFT)** workflows to improve LLM response quality, alignment, and conversational consistency — building strong quantitative evaluation and analytical research skills.
- Conducted systematic **AI data annotation and model output evaluation** against quality benchmarks, applying statistical thinking and pattern recognition in production pipelines.
- Performed **prompt engineering and optimization** to enhance model reasoning across diverse task types, requiring deep analytical thinking and data-driven iteration.
- Contributed to **training dataset curation and research** on emerging LLM alignment techniques under strict data privacy and QA standards.

Full Stack Developer Intern Jan 2026 – Jun 2026

Bluestock Fintech (Remote)

- Shipped production-ready React components for **Logic Looper** (logic-looper-nine.vercel.app), translating user engagement data into interactive visual systems — gaining exposure to fintech data workflows.
- Architected **real-time data synchronization** between IndexedDB and PostgreSQL for persistent user progress tracking, demonstrating strong backend data management and structured data thinking.
- Implemented a **GitHub-style activity heatmap and achievement system** to drive daily user retention, applying data visualization and engagement analytics.

Data Analyst Intern Jun 2025 – Jul 2025

Elevate Labs (Remote)

- Performed **in-depth quantitative data analysis** on large datasets using Python (Pandas, NumPy) and Excel, extracting actionable business insights from unstructured data — awarded **Best Performer** for the cohort.
- Built **interactive Power BI dashboards** to visualize KPIs and designed data stories for non-technical stakeholders, demonstrating strong market analysis and reporting capabilities.
- Applied **statistical analysis, data cleaning, and feature interpretation** techniques to curate and report on complex datasets — directly applicable to quantitative research workflows.

Project Experience

Consumer Segmentation & Quantitative Analytics Dashboard | *Django · Scikit-Learn · RFM · K-Means · Python*2024

- Engineered a **quantitative segmentation engine** automating RFM analysis and K-Means clustering across **500K+ transactions**, uncovering **\$8.9M in revenue opportunities** — directly analogous to market analysis and quant research workflows.
- Built **churn prediction pipeline** identifying **1,300+ at-risk users** using statistical modeling and ML, demonstrating risk assessment and pattern recognition skills applicable to risk management.
- Applied **time-series and statistical analysis** techniques to derive actionable insights from large financial-scale datasets.

Energy Consumption Forecasting | *Python · LSTM · Time-Series · Power BI*2024

- Forecasted industrial energy usage using **LSTM time-series models**, achieving **MAPE of 8.5%** — demonstrating hands-on experience with time-series forecasting directly relevant to financial data modeling.
- Built **interactive Power BI dashboards** for scenario analysis and peak load simulation, enabling data-driven decision-making.

Predictive Maintenance using Sensor Data | *Python · Scikit-Learn · Random Forest · Pandas*2024

- Processed **NASA Turbofan engine dataset** to predict machine failure cycles; engineered features and trained Random Forest achieving **87% accuracy**.
- Designed dashboards to visualize machine health metrics — demonstrating end-to-end ML pipeline development and model evaluation capabilities.

RAG Document Search Pipeline | *FastAPI · LangChain · HuggingFace · ChromaDB · Streamlit*2024

- Built a modular **AI pipeline** processing 50+ documents/min with <200ms latency using RAG architecture — demonstrating ability to build and optimize production-grade ML systems at scale.
- Integrated **HuggingFace Transformers** and ChromaDB for embedding generation and retrieval; applied prompt engineering and model evaluation to optimize output quality.

Education

Madhav Institute of Technology and Science, Gwalior2022 – 2026

B.Tech in Internet of Things — CGPA: 7.5

Relevant Coursework: Machine Learning, Data Science, Statistics, Optimization, Data Structures, SQL

Leadership & Activities

Social Media Head & Core Member | Science Club, MITS2022 – 2026

Led digital outreach and branding campaigns, **increasing engagement by 60%**. Collaborated with cross-functional teams on communications and data-driven content strategy.

Graphic Designer | Science Club, MITS2023 – 2025

Designed visual content and infographics supporting educational outreach for underprivileged communities.

College Football Team | MITS

Represented the institute in nodal-level tournaments, demonstrating teamwork and discipline.

Core Competencies

Machine Learning — Quantitative Research — Statistical Analysis — Time-Series Analysis — Predictive Modeling — Risk Assessment — Data-Driven Research — Python — RLHF & SFT — RAG Pipelines — Feature Engineering — Attention to Detail — Remote Collaboration — Strong English Communication

Certifications

Power BI Virtual Internship — PwC ForageJul 2024

Introduction to SQL — SimplilearnMar 2025

R Programming — SimplilearnJan 2025

Python Basics — Infosys SpringboardMar 2023